

Class Test -III (FEB. - 2023)
AMR2C1- Applied Mathematics -II
(ETC A, B)

Time: 70 min.

Maximum Marks: 20

Note: Attempt all questions. Questions must be solved at one place. Each step should be well defined.

- Q.1 States and prove Baye's Theorem. 5
- Q.2 An anti-aircraft gun can take a maximum of four shots on enemy's plane moving from it. The probabilities of hitting the plane at first, second, third and fourth shots are 0.4, 0.3, 0.2 and 0.1 respectively. Find the probability that the gun hits the plane. 5
- Q.3 Solve the equations by Cardan's Method $x^3 - 6x^2 + 6x - 5 = 0$ 5
- Q.4 Let $X = \{x_1, x_2, x_3, x_4, \}$ and two fuzzy sets A and B are 5
- $A = \{(x_1, 0.2), (x_2, 0.5), (x_3, 0.7), (x_4, 1)\}$
- $B = \{(x_1, 0.6), (x_2, 1), (x_3, 0.4), (x_4, 0.3)\}$ then find $A \cup B$ and $A \cap B$. Is A is subset of B .